

Math 421/510: Homework 3

1. (Folland Ex. 5.63) Let $\mathcal{H}$ be an infinite-dimensional Hilbert space. Show that:

- (a) every orthonormal sequence in $\mathcal{H}$ converges weakly to 0;
- (b) every element of the (norm) closed unit ball

$$B = \{x \in \mathcal{H} : \|x\| \leq 1\}$$

is the weak limit of a sequence in the unit sphere

$$S = \{x \in \mathcal{H} : \|x\| = 1\}$$

(hence S is dense in B in the weak topology).

Solution.

- (a) Let $\{e_n\}_{n \in \mathbb{N}}$ be a sequence of orthonormal vectors. Recall that, $e_n \rightarrow 0$ in the weak topology iff $\forall f \in \mathcal{H}^*, f(e_n) \rightarrow f(0) = 0$. Let $f \in \mathcal{H}^*$. Then, by Riesz representation theorem, $f(\cdot) = \langle \cdot, y \rangle$ for some $y \in \mathcal{H}$. But then, by Bessel's inequality, we have that $\sum_{n=1}^{\infty} |\langle y, e_n \rangle|^2 \leq \|y\|^2 < \infty$.

Let $\varepsilon > 0$. Then, there exists some N such that $\sum_{n=N}^{\infty} |\langle y, e_n \rangle|^2 < \varepsilon^2$. However, this implies that for $n > N$,

$$|f(e_n) - 0|^2 = |\langle e_n, y \rangle|^2 = |\langle y, e_n \rangle|^2 \leq \sum_{n=N}^{\infty} |\langle y, e_n \rangle|^2 < \varepsilon^2,$$

so that $|f(e_n)| < \varepsilon$. Therefore, $f(e_n) \rightarrow f(0)$, as desired. Since f was arbitrary, it follows that $e_n \rightarrow 0$ weakly in $\mathcal{H}$.

- (b) If $\|x\| = 1$, then observe that the sequence $(x, x, \dots)$ satisfies our requirement since then $x \in S$. If $\|x\| = 0$, then $x = 0$, so any orthonormal sequence, which we can construct by the Gram-Schmidt process, will converge to $x = 0$ by part a).

Finally, consider the case $0 < \|x\| < 1$. Define $e_1 = x/\|x\|$. Then, $\|e_1\| = 1$. We can then, by the Gram-Schmidt process construct an orthonormal sequence $(e_1, e_2, \dots)$. Now, let $\lambda = \sqrt{1 - \|x\|^2}$. Let $y_n = x + \lambda e_{n+1}$. In particular, since $e_n \perp x$ for $n > 1$, we have that

$$\|y_n\|^2 = \|x + \lambda e_{n+1}\|^2 = \|x\|^2 + \lambda^2 = 1$$

so that $\|y_n\| = 1$. Thus, $y_n \in S$ for $n \in \mathbb{N}$.

To show that $y_n \rightarrow x$ weakly, let $f \in \mathcal{H}^*$. Then,

$$|f(y_n - x)| = |f(\lambda e_{n+1})| = |\lambda| |f(e_{n+1})| \rightarrow 0 \text{ since } \{e_n\}_{n=2}^{\infty} \text{ orthonormal } (f(e_n) \rightarrow^w 0 \text{ by part (a)}).$$

Since f was arbitrary, we conclude $y_n \rightarrow x$ weakly.

2. Show that ℓ^p is separable for $1 \leq p < \infty$ (recall that separable means that it contains a countable dense subset).

Consider the set of sequences $S = \{(q_1, \dots, q_n, 0, \dots) : q_i \in \mathbb{Q}, n \in \mathbb{N}\}$. This set is countable, because for fixed n , it is isomorphic to the product of a finite number of copies of $\mathbb{Q}$, and so it is the union of countably many countable sets, which is countable.

Also, it is in ℓ^p , since for $q \in S$, we have that there exists some N such that for $n > N$ $q_n = 0$, so that

$$\|q\|_p = \sqrt[p]{\sum_{n=1}^N |q_n|^p} < \infty,$$

so that $q \in \ell^p$.

Now, for $x \in \ell^p$ and $\varepsilon > 0$, we have that the balls of the form $B_\varepsilon(x) = \{y : \|x - y\|_p < \varepsilon\}$ form a basis for ℓ^p , so to show that S is dense, we need merely show that each such ball contains an element of S .

Let $x \in \ell^p$ and let $\varepsilon > 0$. Then, since $\|x\|_p < \infty$, there exists some N such that $\sum_{n=N+1}^{\infty} |x_n|^p < \varepsilon^p/2$. Also, we can choose rational numbers $q_1, \dots, q_N$ such that $|q_i - x_i| < \varepsilon/\sqrt[p]{2N}$ for $1 \leq i \leq N$. If we let $q = (q_1, \dots, q_N, 0, \dots)$, then $q \in S$, and

$$\begin{aligned} \|x - q\|_p &= \sqrt[p]{\sum_{n=1}^N |x_n - q_n|^p + \sum_{n=N+1}^{\infty} |x_n|^p} \\ &< \sqrt[p]{N\varepsilon^p/2N + \varepsilon^p/2} = \varepsilon, \end{aligned}$$

as desired.

3. By some measure theory, together with the Weierstrass approximation theorem, one can show that polynomials are dense in $L^p([0, 1])$ (with Lebesgue measure), $1 \leq p < \infty$. Given this, show:
 - (a) $L^p([0, 1])$ is separable;
 - (b) $L^p(\mathbb{R})$ is separable.

Solution.

- (a) Consider the polynomials on $[0, 1]$ with rational coefficients, $S = \mathcal{P}_{\mathbb{Q}}([0, 1])$. Then, $S \cong \cup_{n=1}^{\infty} \mathbb{Q}^n$ (where we construct this isomorphism by associating coefficients with elements in $\mathbb{Q}^n$) so this set is countable. Also, $S \subset L^p([0, 1])$ by assumption, since we are given that the polynomials $\mathcal{P}([0, 1])$ are dense in $L^p([0, 1])$ so they must be members of $L^p([0, 1])$.

Now, S is dense $\Leftrightarrow$ every open ball in $L^p([0, 1])$ contains an element of S . So, let $f \in L^p([0, 1])$, and $\varepsilon > 0$. Then, by assumption, there exists $g \in \mathcal{P}([0, 1])$ such that $\|f - g\|_p < \varepsilon/2$. Let n be the degree of g , so that $g = \sum_{m=0}^n a_m x^m$ and $a_m \in \mathbb{R}$. Then, let $b_m \in \mathbb{Q}$ such that $|a_m - b_m| < \varepsilon/2(n+1)$, and define $h = \sum_{m=0}^n b_m x^m$, so that $h \in \mathcal{P}_{\mathbb{Q}}([0, 1])$. In particular, in $[0, 1]$ we have that $|h - g| \leq \sum_{m=0}^n |b_m - a_m| |x|^m \leq \sum_{m=0}^n |b_m - a_m| < \varepsilon/2$. Then,

$$\begin{aligned} \|h - g\|_p &= \sqrt[p]{\int_{[0,1]} \left| \sum_{m=0}^n (b_m - a_m) x^m \right|^p} \leq \sqrt[p]{\int_{[0,1]} \left| \sum_{m=0}^n |b_m - a_m| \right|^p} \\ &< \sqrt[p]{\int_{[0,1]} \sum_{m=0}^n |\varepsilon/2(n+1)|^p} = \varepsilon/2. \end{aligned}$$

Thus,

$$\|h - f\|_p \leq \|h - g\|_p + \|g - f\|_p < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Therefore, $\mathcal{P}_{\mathbb{Q}}([0, 1])$ is a countable dense subset of $L^p([0, 1])$, so $L^p([0, 1])$ is separable.

- (b) Let $S = \bigcup_{n=1}^{\infty} \{ \sqrt[p]{2n} f(2nx - n) : f(x) \in \mathcal{P}_{\mathbb{Q}}([0, 1]) \}$, where $\mathcal{P}_{\mathbb{Q}}([0, 1])$ are the polynomials with rational coefficients in $[0, 1]$ and 0 in $\mathbb{R} \setminus [0, 1]$. Since this is a countable collection of countable sets it is countable.

Let $g \in L^p(\mathbb{R})$ and $\varepsilon > 0$. We showed in class that $g\chi_{[-n, n]} \rightarrow g$ in $L^p(\mathbb{R})$. Therefore, there exists some n such that $\|g - g\chi_{[-n, n]}\|_p < \varepsilon/2$.

Next, since $g_n = g\chi_{[-n, n]}$ is 0 outside $[-n, n]$, we have that the function $\sqrt[p]{\frac{1}{2n}} g_n\left(\frac{u+n}{2n}\right)$ is zero outside $u \in [0, 1]$, and then we can compute, by change of variables $\frac{u+n}{2n} \rightarrow x$

$$\int_{[0, 1]} \left| \sqrt[p]{\frac{1}{2n}} g_n\left(\frac{u+n}{2n}\right) \right|^p du = \int_{[-n, n]} |g_n(x)|^p dx \leq \int |g|^p < \infty,$$

so $\sqrt[p]{\frac{1}{2n}} g_n\left(\frac{u+n}{2n}\right) \in L^p([0, 1])$.

Then, by part a), there exists some $f \in \mathcal{P}_{\mathbb{Q}}([0, 1])$ such that $\|f(u) - \sqrt[p]{\frac{1}{2n}} g_n\left(\frac{u+n}{2n}\right)\|_p < \varepsilon/2$. In particular, $\sqrt[p]{2n} f(2nx - n) \in S$. This then implies, by change of variables $2nx - n \rightarrow u$, that

$$\begin{aligned} \|g_n(x) - \sqrt[p]{2n} f(2nx - n)\|_p^p &= \int_{[-n, n]} |g_n(x) - \sqrt[p]{2n} f(2nx - n)|^p dx \\ &= \int_{[0, 1]} \left| \sqrt[p]{\frac{1}{2n}} g_n\left(\frac{u+n}{2n}\right) - f(u) \right|^p du \\ &< (\varepsilon/2)^p. \end{aligned}$$

Then, this gives us that:

$$\|g - \sqrt[p]{2n} f(2nx - n)\|_p \leq \|g - g_n\|_p + \|g_n - \sqrt[p]{2n} f(2nx - n)\|_p < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Therefore, S is dense in $L^p(\mathbb{R})$.

4. Show that the sequence $\{e^{ikx}\}_{k=1}^{\infty}$ converges to 0 weakly in $L^p([0, 1])$, $1 < p < \infty$.

(Hint: you may use that any L^p function can be approximated (in the L^p norm) by a simple function whose sets are intervals — a variant of Folland, Thm. 2.26.)

Solution.

We show that, for every $f \in (L^p)^*$, $f(e^{ikx}) \rightarrow f(0) = 0$. Fix $f \in (L^p)^*$. We show that there exists M such that for all $k > M$, $|f(e^{ikx})| < \varepsilon$, so that $f(e^{ikx}) \rightarrow f(0)$.

By the duality theorem, there exists some $g \in L^q$ such that $f(\cdot) = \int_{[0, 1]} (\cdot) g$. We are given that there exists some simple function h whose sets are intervals such that $h \in L^q$ and $\|h - g\|_q < \varepsilon/2$. Then, $h = \sum_{n=1}^N a_n I_n$, where $I_n = (a_n, b_n) \subset [0, 1]$. (Note that these intervals must be finite for h to be in L^q).

Let $M = 4N \max_{1 \leq n \leq N} \{|a_n|\}/\varepsilon$. Then, for $k > M$, we have

$$\left| \int_{[0,1]} e^{ikx} I_n \right| = \left| \int_{a_n}^{b_n} e^{ikx} \right| = \left| \frac{1}{ik} (e^{ikb_n} - e^{ika_n}) \right| \leq \frac{2}{k}.$$

But then, since $k > M$, we have

$$\begin{aligned} \left| \int_{[0,1]} e^{ikx} h \right| &\leq \sum_{n=1}^N |a_n| \left| \int_{[0,1]} e^{ikx} I_n \right| \\ &\leq \sum_{n=1}^N a_n (2/k) < 2N \max_{1 \leq n \leq N} \{|a_n|\} / (4N \max_{1 \leq n \leq N} \{|a_n|\} / \varepsilon) < \varepsilon/2. \end{aligned}$$

But then, we have the following. For all $k > M$,

$$\begin{aligned} \left| \int_{[0,1]} e^{ikx} g \right| &\leq \left| \int_{[0,1]} e^{ikx} (g - h) \right| + \left| \int_{[0,1]} h e^{ikx} \right| \leq \|e^{ikx}\|_p \|g - h\|_q + \varepsilon/2 \\ &= 1 \cdot \|g - h\|_q + \varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Therefore, $\int e^{ikx} g = f(e^{ikx}) \rightarrow f(0)$ and since $f \in (L^p)^*$ was arbitrary it follows that $e^{ikx} \rightarrow 0$ weakly.

5. Let X be a normed vector space, and

$$B = \{x \in X : \|x\| \leq 1\}$$

its (norm) closed unit ball.

(a) Show

$$B = \bigcap_{\substack{f \in X^* \\ \|f\| \leq 1}} \{x \in X : |f(x)| \leq 1\}.$$

(Hint: it may help to use a consequence of the Hahn–Banach theorem.)

(b) Show that B is weakly closed.

(a) ($\rightarrow$) Suppose $x \in B$. Then, for any $f \in X^*$ with $\|f\| \leq 1$, we have that $|f(x)| \leq \|f\| \|x\| = \|x\| \leq 1$, so

$$x \in \bigcap_{\substack{f \in X^* \\ \|f\| \leq 1}} \{x \in X : |f(x)| \leq 1\}.$$

($\leftarrow$) Suppose

$$x \in \bigcap_{\substack{f \in X^* \\ \|f\| \leq 1}} \{x \in X : |f(x)| \leq 1\}.$$

If $x = 0$, then $\|x\| = 0 \leq 1$, so $x \in B$. Assume $x \neq 0$. By a consequence of the Hahn-Banach theorem, there exists some $f \in X^*$ with $\|f\| = 1$ such that $|f(x)| = \|x\|$. Since $f \in X^*$ with $\|f\| \leq 1$, by assumption $|f(x)| \leq 1$. But then, this implies $\|x\| = |f(x)| \leq 1$, so $x \in B$.

- (b) Suppose $x_n \rightarrow x$ weakly in X where $x_n \in B$. If $x = 0$ then since $x \in B$ we are done, so suppose $x \neq 0$. Then, by a consequence of the Hahn-Banach theorem, we have that there exists $f \in X^*$ such that $\|x\| = |f(x)|$. Since $x_n \rightarrow x$ weakly, in particular $|f(x_n) - f(x)| \rightarrow 0$. By the triangle inequality, $\|x\| = |f(x)| \leq |f(x) - f(x_n)| + |f(x_n)| \leq \varepsilon + |f(x_n)|$ for any $\varepsilon > 0$, if we choose n large enough. But since $\|f\| \leq 1$ and $x_n \in B$, we have that $|f(x_n)| \leq 1$. Thus, $\|x\| \leq 1 + \varepsilon$ for all $\varepsilon > 0$, so $\|x\| \leq 1$ and so $x \in B$.

6. Let X be a normed vector space. If $U \subset X$ is open in the weak topology and $x \in U$, then U contains a subset of the form

$$\{y \in X : |f_k(y - x)| < \varepsilon, k = 1, \dots, n\}$$

for some $\varepsilon > 0$ and $f_1, \dots, f_n \in X^*$ (this is a consequence of Folland, Thm. 5.14).

- (a) If X is infinite dimensional, show that any non-empty weakly open set is (norm) unbounded.
- (b) Conclude that the weak topology on an infinite-dimensional normed vector space is strictly weaker than the norm topology.

- (a) Let $U \in \mathbb{T}_w$. To show that U is unbounded, we produce a vector $z \in U$ such that $\|z\| > M$ for any $M \in \mathbb{N}$. Then, as stated, we have that U contains a subset of the form

$$\{y \in X : |f_k(y - x)| < \varepsilon, k = 1, \dots, n\}.$$

Now, consider the set $Y = \bigcap_{k=1}^n \ker(f_k)$. Since each f_i is a linear functional, we have that $\text{codim}(\ker(f_i)) = 1$. Then, $\text{codim}(Y) \leq \sum \text{codim}(\ker(f_i)) = n$. Since X is infinite dimensional, this implies that Y is infinite dimensional. Therefore, there exists some $y \in Y$ such that $y \neq 0$. In particular, $f_k(y) = 0$ for all $1 \leq k \leq n$. Let $\lambda = \frac{M - \|x\|}{\|y\|}$. Then, $\|x + \lambda y\| \geq \|\lambda y\| - \|x\| = \frac{M - \|x\|}{\|y\|} \|y\| - \|x\| = M$. However, we also have that for each f_k ,

$$|f_k((x + \lambda y) - x)| = |f_k(\lambda y)| = 0.$$

Therefore, $x + \lambda y \in U$. Since M was arbitrary, this implies that U is unbounded, since for any M it contains a vector of norm greater than M .

- (b) The topology $\mathbb{T}_w$ is the smallest topology such that every $f \in X^*$ is continuous. Now, fixing $f \in X^*$, the norm implies that $|f(x) - f(y)| \leq \|f\| \|x - y\|$ which implies continuity in the norm topology. Therefore $\mathbb{T}_w$ is weaker than the norm topology. To show that it is strictly weaker, we need to show that the norm topology contains an open set which is not in $\mathbb{T}_w$. Consider the set $B = \{x : \|x\| < 1\}$. Then, B is open in the norm topology, but since it is bounded, and every open set in $\mathbb{T}_w$ is unbounded, $B \notin \mathbb{T}_w$. Thus, $\mathbb{T}_w \subsetneq \mathbb{T}_{\|\cdot\|}$.

7. Consider the sequence $\{x^{(k)}\}_{k=1}^\infty \subset \ell^\infty$ defined by

$$x_n^{(k)} = \begin{cases} 0 & n < k \\ 1 & n \geq k \end{cases}$$

Show that:

- (a) $\|x^{(n)}\|_\infty = 1$ for all n ;
- (b) $x^{(n)} \rightarrow 0$ in the weak-* topology on ℓ^∞ ;
- (c) $x^{(n)}$ has no convergent subsequence in the weak topology on ℓ^∞ .

(Hint: see question 6 on the previous homework.)

Solution.

- (a) $\|x^{(n)}\|_\infty = \sup_j |x_j^{(n)}| = 1$, since $x_j^{(n)} \in \{0, 1\}$, and there exists j (namely $j = n + 1$) such that $x_j^{(n)} = 1$.
- (b) By Theorem 6.15 in Folland (the dual space characterization) we have that $\ell^\infty \cong (\ell^1)^*$. Thus, $x^{(n)}$ converges in the weak-* topology on $\ell^\infty \Leftrightarrow$ for each $y \in \ell^1$, $x^{(n)}(y) = x^{(n)} \cdot y \rightarrow 0$. Let $y \in \ell^1$. Let $\varepsilon > 0$. Then, there exists some N such that $\sum_{j=N}^\infty |y_j| < \varepsilon$. But then, for $n > N$,

$$|x^{(n)} \cdot y| = \left| \sum_{j=n}^\infty y_j \right| \leq \sum_{j=N}^\infty |y_j| < \varepsilon,$$

so $x^{(n)} \cdot y \rightarrow 0$. Since y was arbitrary, we conclude $x^{(n)} \rightarrow 0$ in the weak-* topology.

- (c) Suppose for the sake of contradiction that $x^{(n)} \rightarrow x$ in the weak topology for some $x \in \ell^\infty$. By Proposition 2.16 in the notes, we have for each $y \in \ell^1$, $y(\cdot) = \sum_{j=1}^\infty y_j(\cdot)_j$ is a linear functional in $(\ell^\infty)^*$. In particular, $y^{(k)} = (0, \dots, 0, 1, 0, \dots)$, where the 1 is in the k -th element, is a member of $(\ell^\infty)^*$. Now, we have that $|y^{(k)}(x^{(n)}) - y^{(k)}(x)| = |x_k^{(n)} - x_k| \rightarrow 0$. Since $x_k^{(n)} = 0$ for all $n > k$, this implies that $x_k = 0$.

Now, in Homework 1 question 6, we showed that there exists $T \in \ell^\infty$ such that $T((1, 1, 1, \dots)) = 1$, and if $z = (z_1, z_2, \dots)$ satisfies $\lim_{j \rightarrow \infty} z_k = L$, then $T(z) = L$. Since $x = 0$, $T(x) =$

0. However, $\lim_{j \rightarrow \infty} x_j^{(n)} = 1$ for each $x^{(n)}$. Therefore, $T(x^{(n)}) = 1$ for each $x^{(n)}$, so $|T(x^{(n)}) - T(x)| \rightarrow 1$.

This contradicts that $x^{(n)} \rightarrow x$ in the weak topology on ℓ^∞ .